



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Macrostates and microstates
- Counting microstates
 - Constrained N and total energy: $\Omega(E)$
 - Distinguishable particles: permuting labels
 - Ω increases rapidly with N, E and decreasing level spacing
- Postulates:
 - 1. Equal probability of all microstates in an isolated system.
 - 2. For an isolated system, $S = k \ln \Omega$ (Boltzmann)
 - Second Law implies $\Omega_{\text{final}} > \Omega_{\text{initial}}$

Summary of last lecture

- The Boltzmann (canonical) distribution
 - System thermostatted by a large, isolated heat bath.
 - High-energy states of the system decrease Ω_{bath}
 - $P_m = Q_N^{-1} \exp(-E_m/kT)$
 - No restrictions on the nature of the system so far.
- The partition function
 - Normalisation “constant” of the Boltzmann distribution.
 - $Q_N = \sum_m \exp(-E_m/kT)$
 - Function of temperature: $Q_N(T)$
 - Bridge relation: $A = -kT \ln Q_N$

Thermodynamic functions in terms of partition functions

- Armed with the canonical distribution function we can proceed to find expressions for the thermodynamic quantities of the system. Recall that before we had **$S = k \ln \Omega$** , but that this applied only to an isolated system. We need a new set of relationships for the case of our system in contact with a heat bath.

Internal energy

- The internal energy of the system is just given by

$$U = \sum_m P_m E_m$$

- The argument here is that the probability of the system being in a microstate with energy E_m is P_m , so the average energy is found by summing $P_m E_m$ over all possible microstates of the system.

Inserting the formula for P_m we have

$$U = \frac{1}{Q_N} \sum_m E_m \exp\left(\frac{-E_m}{kT}\right).$$

- This can be simplified somewhat by considering the derivative of Q_N with respect to T at constant volume:

$$\begin{aligned} \left(\frac{\partial Q_N}{\partial T}\right)_V &= \frac{\partial}{\partial T} \sum_m \exp\left(\frac{-E_m}{kT}\right) \\ &= \frac{1}{kT^2} \sum_m E_m \exp\left(\frac{-E_m}{kT}\right). \end{aligned}$$

Internal energy

- Rearranging the last lines gives

$$\sum_m E_m \exp\left(\frac{-E_m}{kT}\right) = kT^2 \left(\frac{\partial Q_N}{\partial T}\right)_V$$

and substituting this into the equation in last slide gives

$$U = kT^2 \frac{1}{Q_N} \left(\frac{\partial Q_N}{\partial T}\right)_V$$

which, as $d(\ln y) / dx = (1/y) dy / dx$, can be written

$$U = kT^2 \left(\frac{\partial \ln Q_N}{\partial T}\right)_V.$$

- This is a typical result from statistical thermodynamics. *A thermodynamic function can be calculated from the partition function or, as in this case, the derivative of the partition function with respect to temperature.* This is a remarkable result; *Q_N is just a dimensionless number, yet knowledge of its value allows us to calculate thermodynamic quantities!*

Helmholtz energy

- We will simply ***assert*** that the Helmholtz energy is calculated using

$$A = -kT \ln Q_N$$

- This is sometimes called *the ‘bridge’ relation as it establishes the bridge between classical and statistical thermodynamics.*

Entropy

- We can use the following equation derived earlier to find an expression for the entropy:

$$\left(\frac{\partial A}{\partial T}\right)_V = -S$$

- so

$$\begin{aligned} S &= -\frac{\partial}{\partial T} (-kT \ln Q_N) \\ &= k \ln Q_N + kT \left(\frac{\partial \ln Q_N}{\partial T}\right)_V \end{aligned}$$

From its definition, Q_N is clearly temperature dependent.

Internal energy (again)

- An expression for $\mathbf{U}$ can be found by using the definition of $\mathbf{A}$ together with the expressions for $\mathbf{S}$ and $\mathbf{A}$:

$$A = U - TS, \text{ hence } U = A + TS$$

$$\begin{aligned} U &= -kT \ln Q_N + T \left[k \ln Q_N + kT \left(\frac{\partial \ln Q_N}{\partial T} \right)_V \right] \\ &= kT^2 \left(\frac{\partial \ln Q_N}{\partial T} \right)_V. \end{aligned}$$

- This is the same as that found, by a somewhat different method. This may be seen as a justification (but not a proof) of the validity of the bridge relationship.
- We see that all we need to know is the relationship between $\mathbf{A}$ and $\mathbf{Q}_N$; the rest of the thermodynamic functions can then be derived from this.

Course outline

- Evaluation of the partition function

Course outline

- So far all we have is some equations relating partition functions to thermodynamic quantities, but these relationships are of little use to us until we have a practical way of evaluating the partition function. In this section we will explore how in one particular case – that of non-interacting particles – it is possible to evaluate the partition function relatively easily by using the results we already know from quantum mechanics and spectroscopy. The result will be a practical set of relationships for calculating thermodynamic functions.

Non-interacting particles

- If the particles (atoms or molecules) in the sample *do not interact with one another to a significant extent, then the energy levels of one particle are not affected by the positions of the other particles in the system. So, the energy levels of the system are just those available to the particles which make up the system.* Gases are, to a good approximation, representative of non-interacting systems; solids and liquids are not.
- Suppose that the system consists of particles A, B, C The energy of the system will depend on which energy level each particle occupies; let particle A occupy level i , particle B occupy level j , particle C occupy level k and so on. The energy of the microstate with this arrangement is simply the sum of the energies of the particles:

$$E_m = \varepsilon_i^{(A)} + \varepsilon_j^{(B)} + \varepsilon_k^{(C)} + \dots$$

where $\varepsilon(A)$ is the energy of the i th energy level of particle A and so on.

Non-interacting particles

- *The partition function, Q_N , is defined as*

$$Q_N = \sum_m \exp\left(\frac{-E_m}{kT}\right)$$

which becomes, for the $\mathbf{E}_m$ given above,

$$\begin{aligned} Q_N &= \sum_{i, j, k} \exp\left(\frac{-\left(\varepsilon_i^{(A)} + \varepsilon_j^{(B)} + \varepsilon_k^{(C)} + \dots\right)}{kT}\right) \\ &= \left[\sum_i \exp\left(\frac{-\varepsilon_i^{(A)}}{kT}\right) \right] \left[\sum_j \exp\left(\frac{-\varepsilon_j^{(B)}}{kT}\right) \right] \left[\sum_k \exp\left(\frac{-\varepsilon_k^{(C)}}{kT}\right) \right] \dots \end{aligned}$$

Particle A can be in any of its energy levels, as can particles B, C . . . , therefore the sum must be carried out over all possible combinations of i, j, k . . .

Non-interacting particles

- Now suppose that particles A, B and C . . . are all the same and so have the same set of energy levels. Each of the square brackets in the previous expression is the same so

$$Q_N = \left[\sum_i \exp \left(\frac{-\varepsilon_i^{(A)}}{kT} \right) \right]^N$$

where N is the number of particles.

- *The quantity in the square bracket is defined as the molecular partition function, q :*

$$q = \sum_i \exp \left(\frac{-\varepsilon_i}{kT} \right)$$

which means that

$$Q_N = q^N. \quad (\text{Distinguishable particles})$$

Non-interacting particles

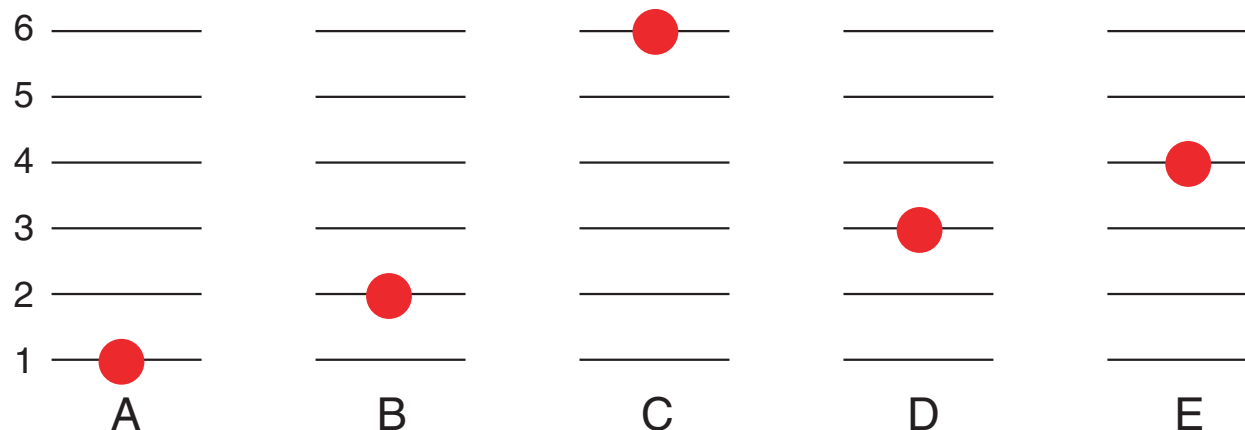
$$Q_N = q^N. \quad (\text{Distinguishable particles})$$

- *The key thing here is that it is possible to know the energy levels of a single molecule* – we can find these from a combination of spectroscopy and quantum mechanics. *Therefore, we can evaluate q and hence Q_N ; then we can compute thermodynamic functions.*
- *Do not confuse q , the molecular partition function with q , the heat*; it should be clear from the context which is meant!
- This equation only applies to ***distinguishable*** particles; it needs to be modified for the case of indistinguishable particles – see the following.

Distinguishable and indistinguishable particles

- The argument which leads to the equation has a problem which results from the fact that it counts all possible arrangements of particles amongst the energy levels i.e. for each level of particle A, terms in the sum are included for all the levels of particle B, all the levels of particle C and so on. If particles A, B and C are the same (indistinguishable), then, for example, A being in level 1 and B being in level 2 is indistinguishable from A being in level 2 and B being in level 1. Under these circumstances this equation will overestimate Q_N .
- To illustrate the point further consider a system with 5 particles, A, B . . . , which each have available to them the same set of energy levels. One arrangement of the particles amongst the energy levels is

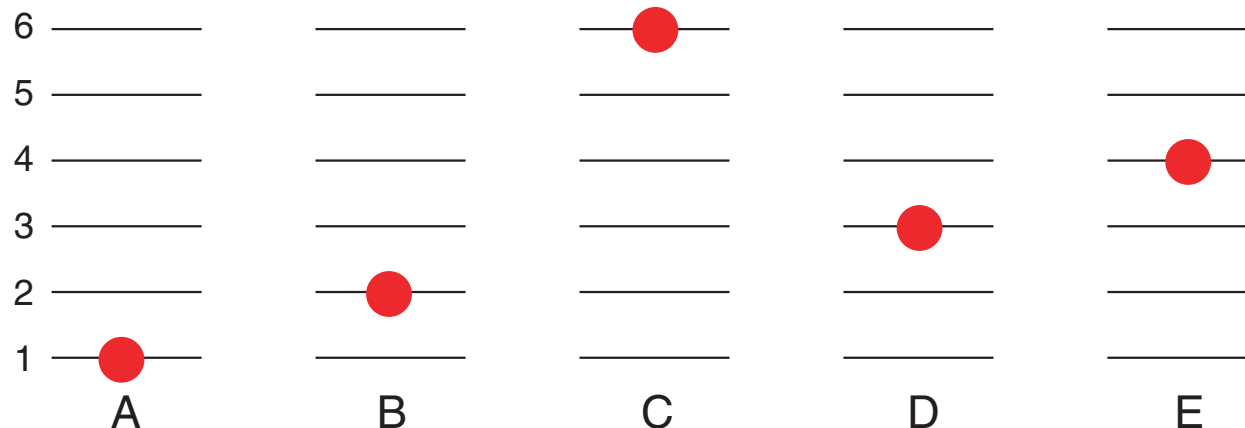
$$Q_N = q^N$$



Distinguishable and indistinguishable particles

- There are many arrangements which have the same occupation of energy levels as this one but which differ according to which particular particle is in each level. For example, particle A could be in level 2 and particle B in level 1. We can work out the number of these arrangements by noting that we have 5 particles to choose from to go in level 1, then 4 to go in level 2, 3 in level 6, 2 in level 3 and finally one in level 4. So the total number of ways is $5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$. Each of these arrangements is included when the partition function, Q_N , is computed according to the equation.
- However, it is clear that if the particles are indistinguishable these 120 arrangements are all the same. Therefore the value of Q_N computed according to this equation is too large by a factor of $5!$

$$Q_N = q^N$$



Distinguishable and indistinguishable particles

- If *there is only one particle per energy level*, Q_N for indistinguishable particles is well approximated by

$$Q_N = \frac{q^N}{N!}. \quad (\text{Indistinguishable particles})$$

- In practice, *because of the very large number of translational energy levels that are accessible, the condition that there be only one particle per energy level is obeyed and the equation above can be used for practical purposes.*
- *Generally molecules and atoms of the same type are indistinguishable, so this equation applies. The exception is in a solid where, although the atoms or molecules themselves are indistinguishable, they are considered to be distinguishable as they occupy lattice sites which are distinguishable.*

Thermodynamic quantities in terms of q

- We can now re-write the expressions for the thermodynamic parameters given earlier in terms of q .
 - Distinguishable particles
 - Indistinguishable particles
- There is no point in remembering all these relationships; *you should be able to work them all out as you need them just from $A = -kT \ln Q_N$, $S = -(\partial A / \partial T)_V$ and $A = U - TS$.*

Distinguishable particles

- In this case $\mathbf{Q}_N = \mathbf{q}^N$. So

$$\begin{aligned} A &= -kT \ln Q_N = -kT \ln q^N \\ &= -NkT \ln q \end{aligned}$$

$$\begin{aligned} S &= k \ln Q_N + kT \left(\frac{\partial \ln Q_N}{\partial T} \right)_V = k \ln q^N + kT \left(\frac{\partial \ln q^N}{\partial T} \right)_V \\ &= Nk \ln q + NkT \left(\frac{\partial \ln q}{\partial T} \right)_V \end{aligned}$$

$$\begin{aligned} U &= kT^2 \left(\frac{\partial \ln Q_N}{\partial T} \right)_V = kT^2 \left(\frac{\partial \ln q^N}{\partial T} \right)_V \\ &= NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V \end{aligned}$$

Indistinguishable particles

- In this case $Q_N = q^N / N!$ The factorial is inconvenient to compute, but as N is very large we can approximate its logarithm to high precision using *Stirling's approximation*:

$$\ln N! = N \ln N - N.$$

- Using this we have:

$$\begin{aligned} A &= -kT \ln \frac{q^N}{N!} \\ &= -NkT \ln q + kT (N \ln N - N) \end{aligned}$$

$$\begin{aligned} S &= k \ln \frac{q^N}{N!} + kT \left(\frac{\partial \ln q^N / N!}{\partial T} \right)_V & U &= kT^2 \left(\frac{\partial \ln q^N / N!}{\partial T} \right)_V \\ &= Nk \ln q - k \ln N! + kT \left(\frac{\partial (N \ln q - \ln N!)}{\partial T} \right)_V & &= kT^2 \left(\frac{\partial (N \ln q - \ln N!)}{\partial T} \right)_V \\ &= Nk \ln q - Nk \ln N + Nk + NkT \left(\frac{\partial \ln q}{\partial T} \right)_V & &= NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V \end{aligned}$$

- Note that *the expression for A and S are different to those for the case of distinguishable particles, whereas the expressions for U are the same in both cases.*

Energy levels

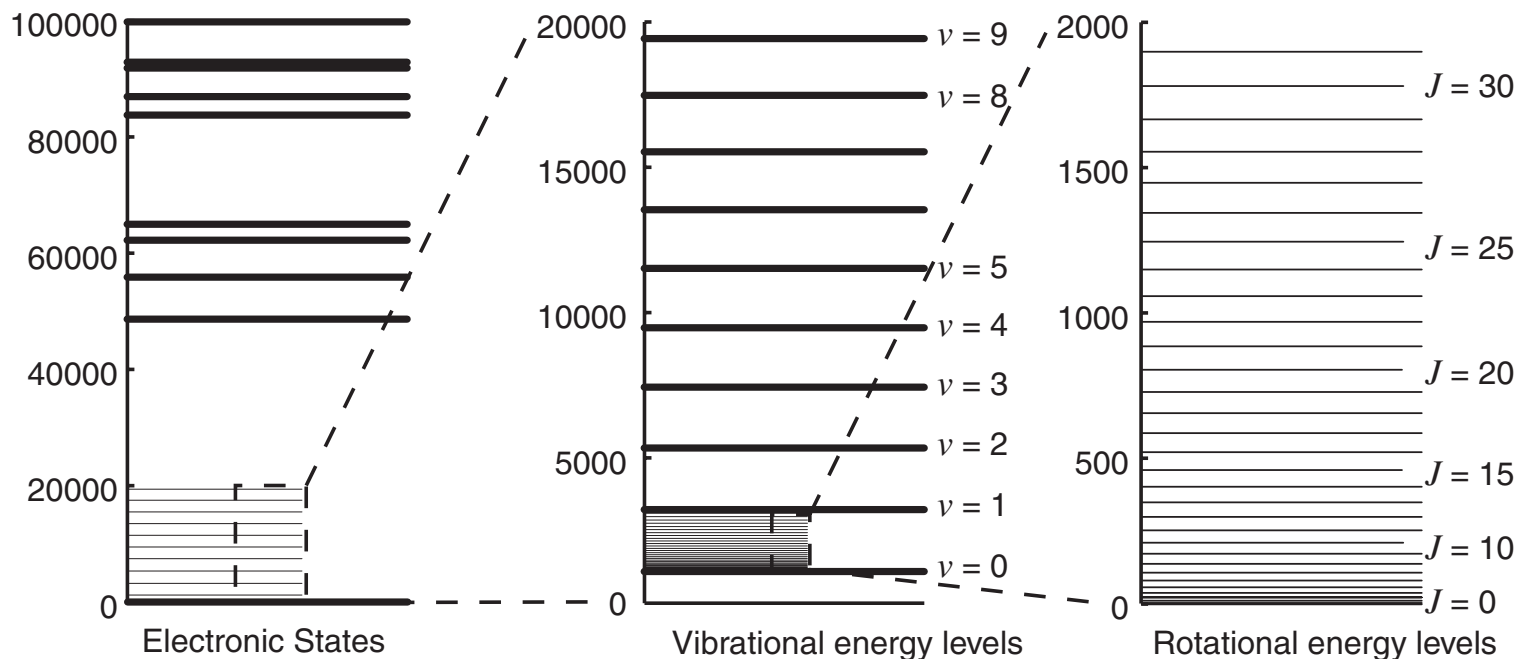
- We see now that the key thing is to be able to evaluate the *molecular partition function*, q . To do this we need to know about the energy levels available to a single atom or molecule – a task which is readily achieved using a combination of quantum mechanics and spectroscopy.
- It is usual to separate the total energy of a molecule or atom into contributions from different kinds of motion (the Born–Oppenheimer separation):
 - **Translational**: this is simply kinetic energy due to the motion of the molecules in space. It can further be separated into contributions from motion in the x, y and z directions.
 - **Rotational**: this is the energy due to overall rotation of a molecule.
 - **Vibrational**: this is the energy due to the vibration of bonds in a molecule.
 - **Electronic**: this is the energy due to the interaction of the electrons and nuclei
- Note that atoms do not possess rotational or vibrational energy levels.

Energy levels

- The total energy can be written as a sum of the different contributions:

$$\mathcal{E}_{\text{tot}} = \mathcal{E}_{\text{trans}} + \mathcal{E}_{\text{rot}} + \mathcal{E}_{\text{vib}} + \mathcal{E}_{\text{elec}}.$$

- The justification for this separation is that the energies are very different in size.
- Each kind of motion has a set of levels associated with it, so we can talk of vibrational energy levels, or translational energy levels and so on.



Different scales of
energy levels for
carbon monoxide

Energy levels

- For each of these set of levels we can define a partition function. For example, for translation we have the translational partition function

q_{trans} :

$$q_{\text{trans}} = \sum_i \exp\left(\frac{-\epsilon_{\text{trans},i}}{kT}\right)$$

where $\epsilon_{\text{trans},i}$ are the translational energy levels; likewise we can define q_{rot} , q_{vib} , and q_{elec} .

- *As the energies are additive, the partition functions are multiplicative*, so the molecular partition function is found from

$$q = q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}.$$

- This form of q implicitly assumes that any energy level due to translation can go with any energy level due to rotation and so on for the other levels. This is virtually always the case (with exceptions).
- *The problem of finding the partition function, q , is reduced to the problem of finding the partition functions for translation, rotation etc.*

Choice of energy zero

- For a given set of energy levels (such as rotation or vibration) *it is usually convenient to express their energies relative to that of the lowest of the levels, i.e. the ground state with index $i = 0$. This choice makes $\epsilon_0 = 0$* and so the first few terms of the partition function are

$$q' = 1 + \exp\left(\frac{-\epsilon_1}{kT}\right) + \exp\left(\frac{-\epsilon_2}{kT}\right) + \dots$$

- *The prime has been added to q to remind us that the energies are measured relative to the ground state.* Other ways of describing this form of the partition function are to say that it is ‘computed assuming that the ground state has zero energy’ or that ‘the energy zero is taken to be the ground state’.

Choice of energy zero

- Of course, in reality the energy of the ground state may not actually be zero but rather have some value ϵ_0 . The energies of the levels which were 0, ϵ_1 , ϵ_2 , . . . now become ϵ_0 , $(\epsilon_1 + \epsilon_0)$, $(\epsilon_2 + \epsilon_0)$, . . . , i.e. all the levels are shifted by ϵ_0 . The partition function computed using these levels is

$$\begin{aligned} q &= \exp\left(\frac{-\epsilon^0}{kT}\right) + \exp\left(\frac{-(\epsilon_1 + \epsilon^0)}{kT}\right) + \exp\left(\frac{-(\epsilon_2 + \epsilon^0)}{kT}\right) + \dots \\ &= \exp\left(\frac{-\epsilon^0}{kT}\right) \left[1 + \exp\left(\frac{-\epsilon_1}{kT}\right) + \exp\left(\frac{-\epsilon_2}{kT}\right) + \dots \right] \\ &= \exp\left(\frac{-\epsilon^0}{kT}\right) \times q'. \end{aligned}$$

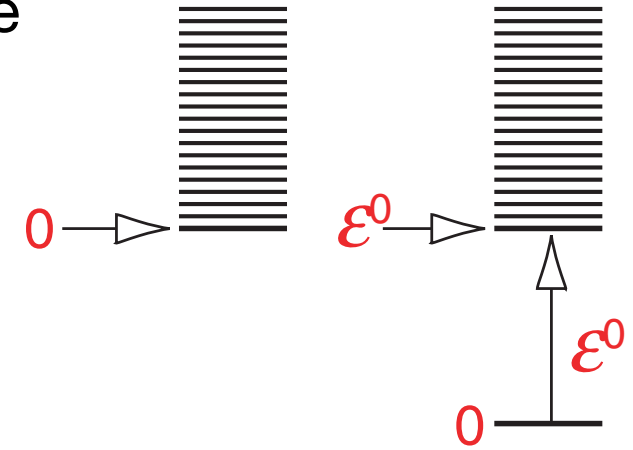
- It is therefore a simple matter *to correct the value of the partition function for a shift in the energy of the ground state to which the energies of the levels are referenced; all we have to do is to multiply by $\exp(-\epsilon_0 / kT)$.*

Effect on the value of the thermodynamic functions

- Shifting the energy of the ground state has an effect on the internal energy. This can easily be seen by inserting the expression for q into the expression for U

$$\begin{aligned}U &= NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V \\&= NkT^2 \frac{\partial}{\partial T} \left(\ln q' - \frac{\varepsilon^0}{kT} \right)_V \\&= NkT^2 \left(\frac{\partial \ln q'}{\partial T} \right)_V + N\varepsilon^0\end{aligned}$$

- Not surprisingly, we see that if *the energy of the ground state is ε_0 then this makes a contribution $N \varepsilon_0$ to the internal energy of N particles*. This term can be identified as the energy that the system has at absolute zero when all the particles are in the ground state; we will denote this energy as U_0 .



On the left is shown a set of energy levels whose ground state has energy zero. At absolute zero, the internal energy is 0 as all particles are in the lowest level. On the right, the energy of the lowest level has been redefined as ε^0 (relative to some other origin). Now the internal energy at absolute zero is $N\varepsilon^0$ for N particles.

Effect on the value of the thermodynamic functions

- Similarly, the value of the Helmholtz energy is affected by this choice of energy zero

$$\begin{aligned} A &= -NkT \ln q + kT (N \ln N - N) \\ &= -NkT \ln q' - NkT \left(\frac{-\varepsilon^0}{kT} \right) + kT (N \ln N - N) \\ &= -NkT \ln q' + kT (N \ln N - N) + N\varepsilon^0 \end{aligned}$$

- In fact *the correction term, $N \varepsilon_0$, is the same as it was for the internal energy; again this term can be identified as the Helmholtz energy at absolute zero.*
- It can be shown that *in the expression for the entropy we simply replace q by q' to give*

$$S = Nk \ln q' - Nk \ln N + Nk + NkT \left(\frac{\partial \ln q'}{\partial T} \right)_V$$

Choice of energy zero in particular cases

- *For translational and rotational energy levels we will see that the ground state really does have energy zero*, and so none of the complications described just now arise. *For vibrational and electronic energy levels it is not the case that the ground state actually has energy zero* and so we will need to take this into account when computing the corresponding thermodynamic functions.

Evaluating q – the importance of kT

- Before we look at particular cases, we will consider in general how we can go about evaluating q for a particular set of energy levels, ϵ_i . Using the definition of q , we can write out a few terms:

$$q = \exp\left(\frac{-\epsilon_0}{kT}\right) + \exp\left(\frac{-\epsilon_1}{kT}\right) + \exp\left(\frac{-\epsilon_2}{kT}\right) + \dots$$

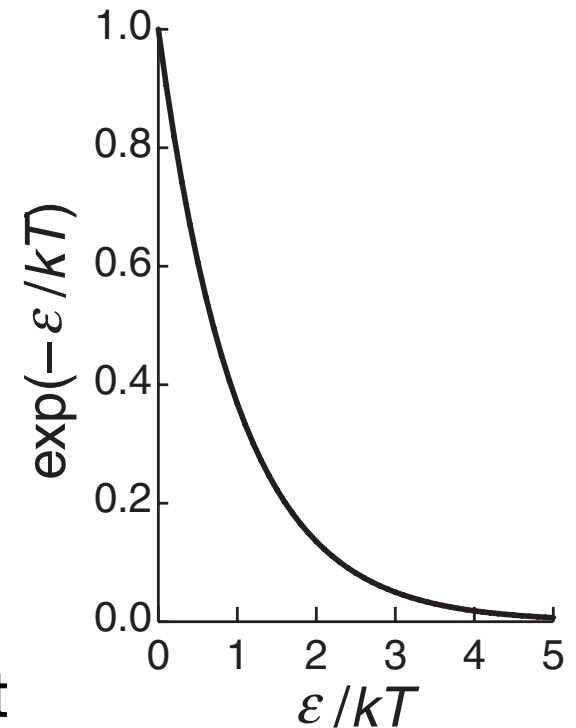
- Assuming that the energies ϵ_i are known, this sum can simply be evaluated term by term. If, as discussed above, we define the energy of the lowest state, the one with index 0, as 0: $\epsilon_0 = 0$, then

$$q' = 1 + \exp\left(\frac{-\epsilon_1}{kT}\right) + \exp\left(\frac{-\epsilon_2}{kT}\right) + \dots$$

- With this definition, *it is clear that the minimum value of q' is 1.*

Evaluating q – the importance of kT

- The question is, how many terms is it necessary to include when computing this sum? This is an important practical question as there are often a very large number of energy levels!
- Due to the properties of exponentials, the second term is less than 1 and the subsequent terms get smaller and smaller as the energy ϵ_i increases. Eventually the terms will be insignificant and need not be included in the sum. Clearly, once ϵ_i exceeds kT by a significant factor, $\exp(-\epsilon_i/kT)$ will be $\ll 1$ and the term will not be significant. In summary:
 - *Only energy levels with energies comparable to or less than kT make a significant contribution to the partition function.*



Evaluating q – the importance of kT

- This is an exceptionally important observation as it affects not only the way in which we evaluate partition functions but it also reveals *the key idea that kT is the scale by which we judge energy levels.* The product kT has dimensions of energy. Over and over again we will use this comparison between the energy of a level and kT .
- Recalling the discussion earlier, the Boltzmann distribution predicts that only those levels whose energies are less than kT will be occupied to a significant extent. So, the levels which are occupied are precisely those which contribute in the evaluation of the partition function.
- Another way of looking at the sum in q is in terms not of the energy levels, but in the spacing of the energy levels. Suppose that the levels are spaced equally by an amount Δ , so that $\epsilon_0 = 0$, $\epsilon_1 = \Delta$, $\epsilon_2 = 2\Delta$, $\epsilon_3 = 3\Delta \dots i\Delta$. The expression of q becomes

$$q' = 1 + \exp\left(\frac{-\Delta}{kT}\right) + \exp\left(\frac{-2\Delta}{kT}\right) + \exp\left(\frac{-3\Delta}{kT}\right) \dots$$

Evaluating q – the importance of kT

- As before, it is clear that once $i\Delta$ becomes significantly greater than kT , the term in the sum will not be significant. There are two limiting cases:
 - *Spacing much greater than kT : ($\Delta \gg kT$) in this case the second term is already insignificant and even more so the third and subsequent terms. The partition function is $= 1$.*
 - *Spacing much less than kT : ($\Delta \ll kT$) in this case many terms will contribute to the sum (up to the term with $i\Delta \approx kT$). The partition function will be significantly greater than 1.*
- In the case that the spacing is comparable to kT , a few terms will contribute and the partition function will be a bit over 1. In summary:
 - *If the energy level spacing significantly exceeds kT , only the ground state term contributes. If the spacing is much less than kT , many terms contribute.*
- From the Boltzmann distribution we see that energy levels which contribute to q , i.e. those with $\epsilon_i \leq kT$, are those levels which will be populated. We can therefore identify q as *the number of accessible states*.